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MASTER THESIS PROPOSAL

Removing Scattered Light from TESS Full Frame Images using a Spatio-Temporal Patch Recurrence Prior

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Declaration of Authorship

I, Shachar Fridman, hereby declare that this thesis proposal entitled, “Removing Scattered Light from TESS Full Frame Images using a Spatio-Temporal Patch Recurrence Prior” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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*“You don’t get to the gold medal without the sweat, the tears, and the urge to quit.
But standing on the podium makes every second worth it.”*

Inspired by Maya Bishop, Station 19

Abstract

The NASA Transiting Exoplanet Survey Satellite (TESS) mission is providing an unprecedented volume of photometric data, crucial for stellar variability, asteroseismology, and exoplanet studies [1]. However, a significant challenge in analyzing TESS Full Frame Images (FFIs) is the presence of instrumental artifacts, most notably scattered light from the Earth and Moon. This non-astrophysical signal can obscure or mimic true stellar variability, complicating large-scale automated analysis.

This work presents a novel, data-driven approach to model and remove scattered light by adapting a computer vision algorithm originally designed for blind image dehazing [2]. We model the scattered light as a "haze" that degrades the underlying "clear" image of the sky. Our methodology is predicated on the principle of internal patch recurrence: the tendency for small patches of an image to repeat within themselves. Scattered light disrupts this self-similarity in a predictable, spatially-varying manner. By identifying these disruptions, we can reverse-engineer the parameters of the scattered light and restore a clean image.

This thesis outlines the development of a 2D spatial framework for this task and details its extension into a 3D spatio-temporal model. This advanced framework leverages the time-series nature of TESS observations, using temporal consistency as a powerful prior to robustly separate the static astrophysical scene from the time-varying instrumental artifact.

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Contents

Declaration of Authorship	i
Abstract	iii
Acknowledgements	iv
1 Introduction	1
1.1 Problem of Interest	1
1.2 Motivation	2
1.2.1 Limitations of Current Astronomical Pipelines	2
1.2.2 Why Internal Patch Recurrence?	2
1.3 Our Contribution	3
2 Related Work	4
2.1 Systematics Correction in Astronomical Surveys	4
2.1.1 Standard Pipeline Approaches	4
2.1.2 Community-Driven and Data-Driven Tools	4
2.2 Deep Learning in Time-Domain Astronomy	5
2.3 Blind Image Dehazing	5
2.4 Spatio-Temporal Consistency	6
3 Methodology	7
3.1 Data Pre-processing: Local Dynamic Range Adaptation	7
3.2 The 2D Spatial Estimation Model	8
3.2.1 Internal Patch Recurrence and Matching	8
Step 1: Removing the Uniform Glow (Mean Subtraction) . . .	9
Step 2: Correcting the Contrast (Normalization)	9
3.2.2 Estimating the Global Scattered Light (A)	9
3.2.3 Recovering the Transmission Map	9
3.3 Global Alignment and 3D Spatio-Temporal Regularization	10
3.3.1 Global Normalization for Physical Consistency	10
3.3.2 Per-Frame Parameter Extraction	10
3.3.3 Temporal Smoothing (Regularization)	11
3.3.4 Final Signal Recovery	11
4 Experimental Design	12
4.1 Datasets	12
4.1.1 Synthetic Dataset: Additive Stray Light Simulation	12
4.1.2 Real-World TESS Dataset	13
4.2 Baseline Methods for Comparison	13
4.3 Evaluation Metrics	13
4.3.1 Image Reconstruction Metrics (Synthetic Data)	14
4.3.2 Photometric Precision (Real Data)	14

5	Results	15
5.1	Quantitative Validation (Simulation Study)	15
5.2	Performance on Real TESS Data	16
5.3	Impact of 3D Spatio-Temporal Regularization	16
6	Discussion	18
6.1	Interpretation of Results	18
6.2	Comparison with Existing Methods	18
6.3	Scientific Implications	19
6.4	Limitations	19
7	Conclusion and Future Work	20
7.1	Summary of Contributions	20
7.2	Future Work	20
7.2.1	Computational Optimization	20
7.2.2	Handling Crowded Fields	21
7.2.3	Deep Learning Integration	21
7.3	Final Remarks	21
	Bibliography	22

List of Figures

- 3.1 **Visual demonstration of Internal Patch Recurrence in TESS data.** The figure displays a full-frame TESS image (center) with two extracted patches, P_1 and P_2 (left panels). Both patches capture the same underlying stellar object but appear different in intensity and contrast due to the spatially varying scattered light. Our algorithm identifies these co-occurring pairs to estimate the scattering parameters. 8
- 5.1 **Dehazing Results on Real TESS Data (Sector 3). Top Row:** The original raw frames, each suffering from strong, non-uniform scattered light gradients that obscure faint stars. **Bottom Row:** The corresponding corrected frames using our algorithm. The background gradients are effectively flattened across all samples, and the contrast of faint stellar targets is significantly enhanced, without introducing ringing artifacts. 16
- 5.2 **Ablation Study: 2D vs. 3D Processing on Real Data. Left:** The original input frame. **Middle (2D Only):** Results obtained from independent frame-by-frame estimation. Note the residual grainy noise and slight intensity inconsistencies in the background, caused by lack of temporal constraints. **Right (3D Regularized):** Results from our proposed spatio-temporal framework. The background is visibly smoother and more consistent. By leveraging the temporal continuity of the orbit, the 3D method effectively suppresses high-frequency estimation noise, yielding a stable photometric baseline. 17

List of Tables

5.1	Quantitative Metrics on Synthetic Dataset	15
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List of Abbreviations

CBV	Co-trending Basis Vectors
CCD	Charge-Coupled Device
CDPP	Combined Differential Photometric Precision
CNN	Convolutional Neural Network
CPM	Causal Pixel Model
CUDA	Compute Unified Device Architecture
CV	Computer Vision
DCP	Dark Channel Prior
DL	Deep Learning
FFI	Full Frame Images
FITS	Flexible Image Transport System
GPU	Graphics Processing Unit
HDR	High Dynamic Range
HEO	High Earth Orbit
ICA	Independent Component Analysis
IPR	Internal Patch Recurrence
MAP	Maximum A Posteriori
ML	Machine Learning
MSE	Mean Squared Error
NASA	National Aeronautics and Space Administration
NL-Means	Non-Local Means
NN	Nearest Neighbors
PCA	Principal Component Analysis
PDC	Pre-search Data Conditioning
ppm	Parts Per Million
PSF	Point Spread Function
PSNR	Peak Signal-to-Noise Ratio
RNN	Recurrent Neural Network
SNR	Signal-to-Noise Ratio
SPOC	Science Processing Operations Center
SSD	Sum of Squared Differences
SSIM	Structural Similarity Index
TESS	Transiting Exoplanet Survey Satellite

List of Symbols

Coordinates and Domains

$\mathbf{x}$	Spatial coordinate vector (x, y) in the image domain
Ω	The spatial domain of the image lattice
t, τ	Temporal index or time variable
N	Total number of frames in the analyzed sequence

Image Formation Model (Haze Analogy)

$I(\mathbf{x})$	The observed (hazy) image; the raw TESS FFI
$L(\mathbf{x})$	The latent (clean) scene radiance
$\hat{L}(\mathbf{x})$	The recovered (dehazed) scene radiance
A	The global scattered light intensity (Airlight)
$\hat{A}$	The estimated airlight value
$t(\mathbf{x})$	The transmission map ($0 \leq t \leq 1$)
$B(\mathbf{x})$	The additive background term, $A(1 - t(\mathbf{x}))$

Simulation Terms

I_{clean}	Ground truth clean image for simulation
I_{syn}	Synthetically generated contaminated image
$S(\mathbf{x}, \tau)$	Simulated additive stray light component
$\eta(\mathbf{x})$	Noise term (photon shot noise and readout noise)

Patch-Based Optimization & Statistics

$\mathcal{P}$	An image patch vector
$\tilde{P}$	Mean-subtracted patch (zero-mean)
$\hat{P}$	Haze-invariant (normalized) patch descriptor
μ_P	Mean intensity of patch P
σ_P	Standard deviation (contrast) of patch P
w	The width of a square patch (e.g., 9×9 pixels)
$d(\cdot, \cdot)$	Distance metric (e.g., Euclidean) in feature space

Spatio-Temporal Regularization

v_{min}, v_{max}	Local (per-frame) minimum and maximum intensity
I_{global}	Globally normalized image cube
$\mathbf{A}_{smooth}(t)$	Temporally smoothed global airlight vector
$\mathbf{T}_{raw}$	The initial 3D transmission volume
$t_{final}(\mathbf{x}, t)$	The final, temporally smoothed transmission map
σ_τ	Standard deviation for temporal Gaussian smoothing

Astronomical Terms

F	Flux (brightness) measured in electrons per second	$[e^-/s]$
T_{mag}	TESS magnitude of a star	$[mag]$
LC	Light Curve (flux as a function of time)	

Chapter 1

Introduction

1.1 Problem of Interest

The **Transiting Exoplanet Survey Satellite (TESS)**, launched in April 2018, marked a significant shift in space-based photometry through its near-all-sky survey strategy. Unlike its predecessor *Kepler*, which continuously monitored a single fixed field of view, TESS employs a survey strategy to cover nearly the entire celestial sphere over the course of its two-year primary mission and subsequent extended missions [1]. The satellite carries four wide-field cameras, each covering a $24^\circ \times 24^\circ$ field of view, creating a combined observational sector of $24^\circ \times 96^\circ$. This optical design allows TESS to monitor millions of bright stars amenable to radial velocity follow-up.

However, the wide-field optical design necessitates a compromise: increased susceptibility to stray light contamination. To maximize sky coverage and maintain a stable thermal environment, TESS operates in a high-Earth elliptical orbit (HEO) in a 2:1 resonance with the Moon. While this orbit is generally stable, it subjects the telescope to complex illumination geometries. Light from the Earth and the Moon can enter the camera lens hoods, scatter off internal optical surfaces, and reach the detectors [3].

The primary problem addressed in this thesis is the contamination of TESS Full Frame Images (FFIs) by this **scattered light**. While community pipelines have been efficient in extracting photometry from these images [4, 5], the complex background remains a major hurdle for faint sources [6, 7]. Unlike uniform background offsets, this artifact manifests as structured, high-amplitude additive noise exhibiting complex spatio-temporal variability.

Spatially, the scattered light manifests as broad, diffuse gradients, intense "straps" of light, or localized glints that can obscure large portions of the sensor. The intensity of this background can sometimes exceed the intrinsic flux of faint target stars by an order of magnitude. Temporally, the background is highly dynamic. As the spacecraft traverses its orbit and the relative angles of the Earth and Moon change, the scattered light patterns shift across the focal plane. These variations occur on timescales ranging from the rapid 2-minute cadence of observations to longer trends spanning the 27-day sector duration [8].

For high-precision photometry, this contamination is critical. The detection of exoplanet transits requires measuring flux dips often smaller than 1% (10,000 ppm), while asteroseismology relies on detecting oscillations with amplitudes of merely a few parts per million (ppm) [9]. If the background level fluctuates, it introduces systematic errors in the aperture photometry. Standard background subtraction methods, which typically estimate the sky level using the median of surrounding pixels, often fail in the presence of scattered light because the background is not locally uniform (it has steep gradients) and the "surrounding" pixels are themselves contaminated.

Furthermore, these background variations often occur on timescales similar to transit durations or stellar pulsation periods, making it difficult to disentangle instrumental noise from astrophysical signals [10]. Even when the background trend is successfully modeled and subtracted, the excess photons increase the photon shot noise, irreversibly degrading the signal-to-noise ratio (SNR) of faint targets. Consequently, uncorrected scattered light often results in unusable data segments, false positives in transit searches, or the masking of real astrophysical variability.

1.2 Motivation

The motivation for this research stems from a critical gap in current data processing pipelines and a unique methodological opportunity to bridge astronomy and computer vision.

1.2.1 Limitations of Current Astronomical Pipelines

Current standard pipelines, such as the SPOC pipeline [11], employ sophisticated background correction methods based on physical instrument models and Co-trending Basis Vectors (CBVs). While highly effective for the majority of the sky, these methods face significant challenges in regions contaminated by complex, high-amplitude scattered light [10].

In such regimes, automated pipelines often flag data as unusable, resulting in significant data loss. Furthermore, community-driven tools that rely on regression techniques (like Principal Component Analysis) carry an inherent risk: they can inadvertently subtract real, long-term astrophysical variability [12]. If a stellar signal (e.g., a supernova curve or rotational period) temporally correlates with the background noise model, regression-based methods may suppress the scientific signal along with the noise [13]. Consequently, high-precision photometry requires a correction methodology independent of global basis set regression.

1.2.2 Why Internal Patch Recurrence?

We address this challenge by adapting ‘Blind Image Dehazing’ techniques from the domain of Computer Vision. However, not all dehazing algorithms are suitable for astronomical data.

The most popular dehazing algorithm, the Dark Channel Prior (DCP) [14], relies on the statistical observation that in daytime natural images (e.g., landscapes), local patches contain pixels with very low intensity in at least one color channel. This assumption fails in astronomical imagery, where the background is inherently dark and the “scene” consists of sparse point sources rather than continuous textures.

Instead, we are motivated by the Internal Patch Recurrence prior proposed by Bahat and Irani [15, 2]. This approach is agnostic to color and relies solely on structural repetition. The astrophysical scene exhibits inherent self-similarity, characterized by the recurrence of Point Spread Functions (PSFs) across the spatial domain. In contrast, scattered light is smooth and non-repetitive at small scales. This specific statistical distinction—which does not exist in standard regression methods—provides a powerful lever to separate the static astrophysical scene from the dynamic instrumental artifact in a fully unsupervised manner.

1.3 Our Contribution

This thesis presents the design, implementation, and evaluation of a novel, unsupervised framework for scattered light removal in TESS FFIs. By bridging the gap between time-domain astronomy and computer vision, we offer a solution that is complementary to existing pipelines. Our specific contributions are as follows:

1. **Adaptation of the Internal Patch Recurrence Prior:** We provide the first application of the Bahat and Irani [2] dehazing model to astronomical data. We rigorously formulate the TESS scattered light problem as a blind source separation task, demonstrating that the statistical differences between the "sparse" stellar field and the "smooth" scattered light are sufficient for unsupervised separation.
2. **Development of a 2D Spatial Solver:** We construct a spatial correction algorithm that processes FFIs frame-by-frame. This module estimates the background map A and transmission $t(x)$ by maximizing the patch recurrence of the recovered scene. This allows for the recovery of visibility in frames that would otherwise be discarded due to high background saturation.
3. **Extension to a 3D Spatio-Temporal Framework:** We introduce a novel 3D spatio-temporal framework, extending the 2D prior to leverage the temporal continuity of TESS observations. TESS data is not a collection of unrelated images, but a continuous video sequence. We introduce a temporal consistency constraint, treating the data as a 3D tensor $(x, y, time)$.

Since the astrophysical scene (star positions) is static over a sector, while the scattered light is dynamic, we enforce that the recovered "clean" image must maximize patch recurrence not just spatially, but across the temporal axis. This 3D approach dramatically reduces flickering artifacts and produces more stable light curves than spatial-only processing.

4. **Simulation-Based Validation Framework:** To overcome the absence of on-orbit ground truth, we design a rigorous validation framework utilizing synthetic data with modeled additive stray light. We generate contaminated datasets by injecting modeled scattered light artifacts into clean TESS frames. This approach allows us to rigorously measure the algorithm's reconstruction fidelity by directly comparing the recovered output against the known, uncontaminated source images.

Chapter 2

Related Work

Our research sits at the intersection of two distinct disciplines: time-domain astronomy and computer vision. While the problem of removing instrumental noise is classically astronomical, our proposed solution is adapted from the domain of computational photography. In this section, we review existing methods for systematic error correction in TESS data, the emerging role of deep learning in light curve analysis, and the relevant background in image dehazing that forms the basis of our approach.

2.1 Systematics Correction in Astronomical Surveys

The removal of instrumental artifacts is a prerequisite for high-precision photometry. In the context of TESS, efforts to mitigate these effects are extensive and can be broadly categorized into pipeline-based approaches and community-driven tools.

2.1.1 Standard Pipeline Approaches

The standard for TESS data reduction is the Science Processing Operations Center (SPOC) pipeline [11], hosted at NASA Ames. SPOC relies on the Pre-Search Data Conditioning (PDC) module [16], which uses a Bayesian Maximum A Posteriori (MAP) approach. This method utilizes Co-trending Basis Vectors (CBVs), common trends derived from an ensemble of quiet stars on the same detector, to model and remove systematic errors.

While highly effective for known systematics (like thermal focus variations), this approach depends heavily on the assumption that the "common mode" of variability is purely instrumental. As noted by Lund et al. [10], in cases of complex, spatially varying scattered light, the global CBVs may not capture local gradients accurately. Conversely, regression-based methods run the risk of over-fitting, potentially suppressing astrophysical signals that typically occur on similar timescales to the noise trends, such as stellar rotation or pulsations.

2.1.2 Community-Driven and Data-Driven Tools

To address the needs of targets not processed by SPOC, several community-developed tools have emerged. The Python package `lightcurve` provides a user-friendly interface for correcting light curves, often employing Principal Component Analysis (PCA) or simple background aperture subtraction. Similarly, `eleanor` [4] is a dedicated pipeline for extracting light curves from TESS Full Frame Images (FFIs), attempting to model systematics by regressing the target light curve against a set of background pixels.

Recent efforts have sought to mitigate the risks of aggressive regression. For example, the Causal Pixel Model (CPM) implemented in the `unpopular` package [12] attempts to preserve long-term astrophysical trends by only using "causal" background pixels that do not contain the target star's signal. However, even these advanced regression methods struggle when the background contamination is non-uniform and highly structured, as is the case with Earthshine scattered light.

2.2 Deep Learning in Time-Domain Astronomy

In recent years, the field has seen a paradigm shift towards machine learning (ML) and deep learning (DL) techniques [17]. Large-scale datasets from Kepler and TESS have enabled the training of sophisticated models for classification and detection.

Naul et al. [18] and Jamal & Bloom [19] demonstrated the efficacy of Recurrent Neural Networks (RNNs) for classifying variable stars from sparse time-series data. More recently, Transformer-based architectures have been applied to capture long-range dependencies in light curves [20]. Specifically addressing the scattered light problem, Muthukrishna et al. [6] and Lupo et al. [7] have proposed generative AI models to clean TESS FFIs.

While these supervised or semi-supervised approaches show great promise, they face a fundamental limitation: the lack of "ground truth." Unlike classification tasks where labels exist, we do not possess a clean version of a TESS image contaminated by scattered light. Training generative models thus requires complex simulations or domain adaptation [21], which may introduce synthetic biases. This motivates our search for an *unsupervised* method that relies solely on the internal statistics of the image.

2.3 Blind Image Dehazing

In computer vision, the restoration of an image degraded by atmospheric scattering is known as "blind image dehazing." Since the transmission map $t(x)$ and the atmospheric light A are unknown, the problem is under-constrained. Solutions therefore rely on strong statistical priors.

Early works like Fattal [22] utilized Independent Component Analysis (ICA) to separate scene albedo from transmission. A breakthrough in the field was the Dark Channel Prior (DCP) proposed by He et al. [14]. The DCP is based on the empirical observation that in non-sky patches of haze-free outdoor images, at least one color channel has very low intensity. However, the DCP is ill-suited for astronomical images. The night sky is predominantly dark (violating the assumption of non-sky patches), and stars are effectively point sources without the rich color texture required for the DCP to distinguish depth.

This limitation necessitates a different prior. Our work is a direct adaptation of the method by Bahat and Irani [2], which introduced the Internal Patch Recurrence prior. This approach generalizes the concept of self-similarity, a cornerstone of image statistics famously utilized in the Non-Local Means (NL-Means) denoising algorithm [23]. Bahat and Irani demonstrated that haze disrupts the natural recurrence of small image patches (5×5 or 7×7 pixels). By identifying patch correspondences that are statistically broken, their algorithm recovers the haze parameters in a fully unsupervised manner.

2.4 Spatio-Temporal Consistency

Finally, our extension of the 2D method to a 3D framework is motivated by advancements in video dehazing. Single-image methods applied frame-by-frame often result in temporal artifacts or flickering. Methods such as those by Kim et al. [24] leverage temporal coherence—the assumption that the underlying scene structure changes slowly—to regularize the transmission map estimation. We adapt this concept to the TESS data cube, using the strict temporal stability of stellar positions as a hard constraint to disentangle the moving scattered light from the static star field.

Chapter 3

Methodology

This research presents a comprehensive computational pipeline for mitigating scattered light artifacts in TESS Full Frame Images (FFIs). By formulating the artifact removal as a blind source separation problem, we adapt the principles of "Blind Dehazing using Internal Patch Recurrence," originally proposed by Bahat and Irani [2] for terrestrial photography, and extend them into a robust 3D spatio-temporal framework tailored for astronomical time-series data.

The methodology is implemented as a multi-stage pipeline comprising local dynamic range adaptation, frame-independent spatial estimation, global alignment, and temporal regularization. The following sections detail the derivation and mathematical formulation of each stage.

3.1 Data Pre-processing: Local Dynamic Range Adaptation

The raw data provided by the TESS mission consists of photon flux counts (electrons per second), represented as 32-bit floating-point numbers [1]. Unlike standard computer vision imagery defined on $[0, 1]$, TESS data possesses a High Dynamic Range (HDR) with no standardized upper bound. Furthermore, the "Internal Patch Recurrence" prior relies on capturing relative contrast variations, which can be obscured if the dynamic range is not handled consistently.

To maximize the efficacy of the 2D spatial solver, we employ a normalization scheme as a preliminary step. For a given input frame $I(x, y)$, we first compute the local extrema:

$$v_{min} = \min_{x,y} \{I(x, y)\} \quad (3.1)$$

$$v_{max} = \max_{x,y} \{I(x, y)\} \quad (3.2)$$

The frame is then linearly stretched to the full unit interval:

$$I_{local}(x, y) = \frac{I(x, y) - v_{min}}{v_{max} - v_{min}} \quad (3.3)$$

This transformation ensures that the frame maximizes its local contrast, providing optimal input conditions for the patch-matching mechanism described in the following section. Note that this scaling is temporary; the normalization constants are stored to restore physical consistency in later stages.

3.2 The 2D Spatial Estimation Model

The core of our algorithm is the estimation of the scattered light component from individual FFI frames. We adopt the optical model for atmospheric scattering, modeling the observed intensity $I(x)$ as a superposition of the true scene radiance $L(x)$ (the star field) and the scattered light [25]:

$$I(x) = L(x)t(x) + A(1 - t(x)) \quad (3.4)$$

Here, A represents the global intensity of the scattered light ("airlight"), and $t(x) \in [0, 1]$ is the transmission map, which describes the portion of light that reaches the camera without being scattered. In regions where the scattered light is dense, $t(x)$ approaches 0, and the pixel value approaches A .

Since Equation 3.4 is under-constrained (containing more unknowns than equations), we rely on a strong statistical prior known as **Internal Patch Recurrence** [2].

3.2.1 Internal Patch Recurrence and Matching

The fundamental assumption is that small image patches (e.g., stellar PSFs) recur at multiple locations within the same image. However, scattered light acts like a spatially varying fog, causing identical stellar patterns to appear different depending on their location in the haze.

To identify "co-occurring patches"—distinct image patches that share the same underlying stellar structure but differ due to haze—we employ a rigorous selection and matching process. Figure 3.1 visually demonstrates this principle.

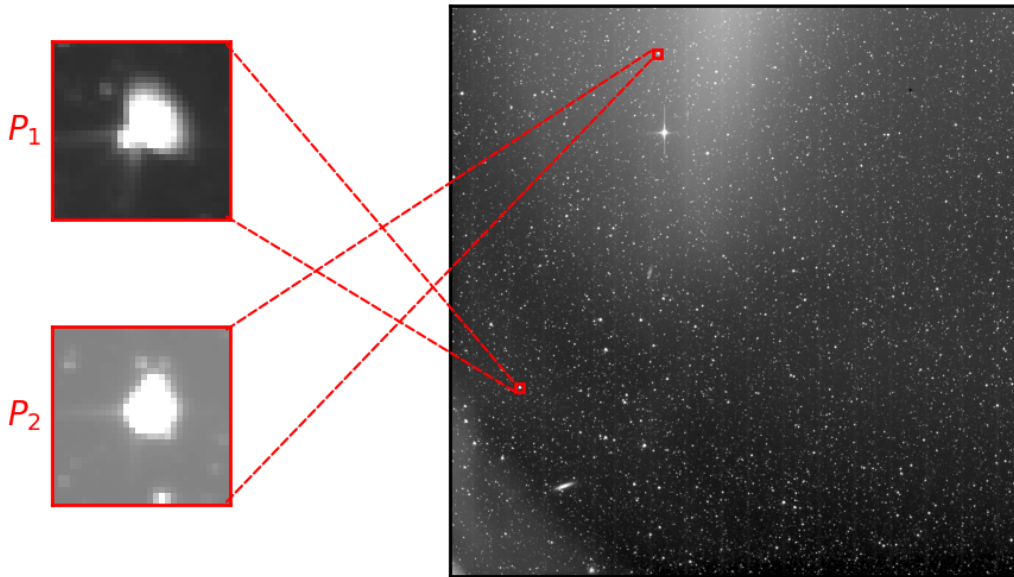


FIGURE 3.1: **Visual demonstration of Internal Patch Recurrence in TESS data.** The figure displays a full-frame TESS image (center) with two extracted patches, P_1 and P_2 (left panels). Both patches capture the same underlying stellar object but appear different in intensity and contrast due to the spatially varying scattered light. Our algorithm identifies these co-occurring pairs to estimate the scattering parameters.

To mathematically identify these pairs despite the interference of haze, we apply a two-step transformation to make the patches invariant to the haze parameters:

Step 1: Removing the Uniform Glow (Mean Subtraction)

The scattered light component A adds a uniform layer of brightness to the patch. Since A is a low-frequency signal, it is approximately constant within the small window of a patch. By subtracting the mean intensity μ_P from each patch, we remove this additive bias:

$$\tilde{P}(x) = P(x) - \mu_P \approx t \cdot (L(x) - \mu_L) \quad (3.5)$$

At this stage, the patch $\tilde{P}(x)$ is free of the additive airlight term, but its contrast is still faded by the unknown transmission scaling factor t .

Step 2: Correcting the Contrast (Normalization)

The next challenge is that one patch is still a scaled version of the other because their transmission values differ ($t_1 \neq t_2$). The standard deviation of a patch is a direct measure of its contrast. To eliminate this scalar dependence, we normalize each mean-subtracted patch by its standard deviation, σ_P :

$$\hat{P}(x) = \frac{\tilde{P}(x)}{\sigma_P} \approx \frac{L(x) - \mu_L}{\sigma_L} \quad (3.6)$$

This normalized representation $\hat{P}$ allows us to compare patches directly. We employ a KD-Tree algorithm to efficiently search the entire image for Nearest Neighbors (NN) in this feature space. Pairs of patches with high normalized correlation are flagged as co-occurring pairs.

3.2.2 Estimating the Global Scattered Light (A)

Each identified pair of co-occurring patches (P_1, P_2) provides a mathematical constraint on the global scattered light value A . Since both patches share the same underlying structure but differ in contrast and offset, they are linearly related.

From Equation 3.4, we derive the following constraint for every pixel x in the patch pair:

$$P_2(x) - A = \frac{t_2}{t_1}(P_1(x) - A) \quad (3.7)$$

where the ratio of transmissions is estimated directly from the patch statistics: $\frac{t_2}{t_1} \approx \frac{\sigma_{P_2}}{\sigma_{P_1}}$. We solve this over-determined system using Least Squares optimization to find a pairwise estimate for A . To determine the global $\hat{A}$ for the entire frame, we compute a weighted average of all pairwise estimates, assigning higher weights to pairs with larger transmission differences.

3.2.3 Recovering the Transmission Map

With the global scattered light $\hat{A}$ determined, we estimate the transmission map $t(x)$ using the physical constraint that the observed intensity cannot be lower than the scattered light component itself ($I(x) \geq A(1 - t)$). This yields a "lower bound" estimate:

$$t_{raw}(x) = 1 - \frac{I(x)}{\hat{A}} \quad (3.8)$$

This initial map contains noise and artifacts. To produce a pixel-accurate map, we refine t_{raw} using a Guided Filter [26]. This advanced smoothing operator uses the original image $I(x)$ as a guidance map, ensuring that the transmission map is smooth in background regions while preserving the sharp edges of stellar profiles.

3.3 Global Alignment and 3D Spatio-Temporal Regularization

The 2D spatial model described above operates on each frame independently. While effective at identifying scattered light structure, treating a TESS observation sector as a sequence of unconnected images introduces a significant challenge: temporal flickering. Small, independent errors in the estimation of $\hat{A}$ or $t(x)$ between consecutive frames can introduce high-frequency noise into the light curves.

To overcome this, we extend the method into a 3D spatio-temporal framework. We leverage the physical nature of the data: the TESS spacecraft orbit is stable, and the scattered light artifacts evolve smoothly over time.

3.3.1 Global Normalization for Physical Consistency

To ensure physical consistency across time and allow for valid temporal constraints, we must align all frames to a common scale. While the local adaptation described previously optimizes contrast per frame, it destroys the relative radiometric relationships between different time steps.

Therefore, we employ a **Global Normalization** scheme. Instead of computing extrema for each frame individually, we calculate the global minimum and maximum intensity values across the entire video sequence simultaneously ($v_{min,global}, v_{max,global}$). The normalized intensity for any pixel at any time t is then given by:

$$I_{global}(x, y, t) = \frac{I(x, y, t) - v_{min,global}}{v_{max,global} - v_{min,global}} \quad (3.9)$$

By using these fixed global bounds, a specific normalized value represents the same physical flux magnitude throughout the entire sequence, preserving temporal coherence.

3.3.2 Per-Frame Parameter Extraction

With the physical consistency established via global normalization, we proceed to extract the scattering parameters for the entire sequence. We apply the 2D spatial estimation model (detailed in Section 3.2) to each time-step t of the normalized data cube $I_{global}(x, y, t)$ independently.

For every frame, the algorithm identifies co-occurring patches and solves for the instantaneous airlight value, yielding a raw time-series vector $\mathbf{A}_{raw}(t)$. Simultaneously, an initial transmission map is computed for each frame, resulting in a raw 3D transmission volume $\mathbf{T}_{raw}(x, y, t)$. At this stage, these estimates are calculated frame-by-frame and capture the instantaneous geometry of the scattered light. However, because they are derived independently, they may exhibit high-frequency variations due to sensor noise or stellar variability, which necessitates the temporal regularization described in the next section.

3.3.3 Temporal Smoothing (Regularization)

With the data aligned, we proceed to filter out the high-frequency "flicker". We apply independent Gaussian smoothing to both the global airlight intensity and the transmission maps.

1. Smoothing the Airlight Intensity (A): We assume that the global intensity of the scattered light changes smoothly. We apply a 1D Gaussian smoothing filter along the temporal axis of the vector $\mathbf{A}_{aligned}$:

$$A_{smooth}(t) = \text{GaussianFilter1D}(A_{aligned}(t), \sigma = 1.5) \quad (3.10)$$

This step effectively filters out outlier estimates caused by frames with low signal-to-noise ratios while preserving the overall trend of the background illumination.

2. Volumetric Smoothing of Transmission Maps (T): The geometric distribution of the scattered light (the shape of the gradients) also evolves slowly. We treat the stack of transmission maps as a 3D volume $\mathbf{T}(x, y, t)$ and apply a Gaussian filter along the temporal axis for every pixel (x, y) :

$$t_{final}(x, y, t) = \text{GaussianFilter1D}(\mathbf{T}_{raw}(x, y, t), \sigma = 1.5) \quad (3.11)$$

This volumetric smoothing enforces temporal consistency, ensuring that the background model does not jump erratically between frames.

3.3.4 Final Signal Recovery

Finally, we recover the clean astrophysical scene $\hat{L}$ for the entire data cube. We invert the haze model using the globally normalized input data I_{global} and the temporally stabilized parameters:

$$\hat{L}_{norm}(x, y, t) = \frac{I_{global}(x, y, t) - A_{smooth}(t)}{t_{final}(x, y, t)} + A_{smooth}(t) \quad (3.12)$$

The resulting cube is then de-normalized to restore the original physical units, yielding a corrected FFI sequence ready for photometric extraction.

Chapter 4

Experimental Design

Following the derivation of the spatio-temporal dehazing framework, this chapter outlines the experimental setup used to validate our methodology. A fundamental challenge in removing scattered light from astronomical imagery is the absence of "ground truth" data; the true astrophysical scene, unobstructed by Earth or Moon scattered light, cannot be observed simultaneously with the artifacts.

To address this, our validation strategy employs a dual approach: a quantitative evaluation using a physics-based simulation with known parameters, and a qualitative and photometric assessment using real-world TESS observations affected by severe scattered light.

4.1 Datasets

We constructed two distinct datasets to evaluate different aspects of the algorithm's performance.

4.1.1 Synthetic Dataset: Additive Stray Light Simulation

To strictly quantify the reconstruction accuracy, we generated a synthetic dataset where the clean astrophysical signal is known. Unlike terrestrial dehazing, where haze causes attenuation, scattered light in space is primarily an **additive** noise source.

We selected a "clean" TESS FFI frame (I_{clean}) from a sector with minimal scattered light contamination to serve as the Ground Truth. To simulate the complex, time-varying nature of scattered light in TESS, we generated a sequence of $N = 10$ frames, $I_{syn}(\mathbf{x}, \tau)$, using an additive formation model:

$$I_{syn}(\mathbf{x}, \tau) = I_{clean}(\mathbf{x}) + S(\mathbf{x}, \tau) + \eta(\mathbf{x}) \quad (4.1)$$

Where:

- $I_{clean}(\mathbf{x})$ is the base stellar field (containing stars and natural background).
- $S(\mathbf{x}, \tau)$ is the simulated time-varying scattered light (Stray Light).
- $\eta(\mathbf{x})$ represents photon shot noise and readout noise.

Simulation of Amorphous Scattered Light: Instead of a simple geometric gradient, we modeled $S(\mathbf{x}, \tau)$ as a dynamic "amorphous cloud" to challenge the algorithm. The stray light structure was generated using Coherent Noise (heavily smoothed Gaussian noise) modulated by a moving spatial envelope. Key simulation parameters included:

1. **Texture:** A non-uniform, clumpy texture generated by Gaussian filtering ($\sigma = 30$) of random noise, mimicking the complex reflection patterns often seen in TESS stray light straps.
2. **Dynamics:** The stray light structure was made to expand and shift across the frame over time, simulating the changing angle of the Earth/Moon relative to the spacecraft’s bore-sight.
3. **Intensity:** To replicate severe contamination, the peak intensity of the simulated stray light was set to approximately $20\times$ the median background level of the clean image.

This dataset allows us to measure pixel-wise error metrics (MSE, PSNR) and structural fidelity (SSIM) which are impossible to calculate on real data.

4.1.2 Real-World TESS Dataset

For astrophysical validation, we targeted TESS sectors known for high levels of scattered light contamination. Specifically, we utilized data from Sector 3, where light from the Earth and Moon frequently impinges on the camera lens hood.

Within these sector, we identified two types of regions for analysis:

1. **Crowded Fields:** Regions near the Galactic Plane with high stellar density [27], used to test the algorithm’s ability to distinguish between dense star clusters and background glow without removing astrophysical signal.
2. **Variable Star Targets:** Specific targets with known variability periods. Preserving the light curves of these stars is critical; a successful correction must remove the background trend without dampening the transit signals or stellar pulsations.

4.2 Baseline Methods for Comparison

To demonstrate the efficacy of our proposed 3D spatio-temporal framework, we compare our results against three baseline categories:

1. **Standard Background Subtraction:** A conventional astronomical approach involving 2D polynomial background fitting. This represents the standard “non-dehazing” approach to background removal often used in quick-look pipelines.
2. **Ablation Study (2D vs. 3D):** To isolate the contribution of our temporal regularization, we compare our full 3D pipeline against a **2D Frame-by-Frame** implementation of the same algorithm (similar to the original Bahat & Irani method [2]). In the 2D baseline, the airlight (A) and transmission (t) are estimated independently for each frame without temporal smoothing.

4.3 Evaluation Metrics

Performance evaluation requires a combination of computer vision metrics (for image quality) and astronomical metrics (for scientific utility).

4.3.1 Image Reconstruction Metrics (Synthetic Data)

For the synthetic dataset, where Ground Truth is available, we employ standard image quality metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between the estimated scene $\hat{L}$ and the ground truth L . Lower is better.
- **Peak Signal-to-Noise Ratio (PSNR):** Evaluates the reconstruction quality in decibels (dB). A value above 40 dB typically indicates excellent restoration.
- **Structural Similarity Index (SSIM):** A perceptual metric [28] that quantifies the degradation of structural information rather than absolute pixel differences. In astronomy, maintaining the shape of the Point Spread Function (PSF) is critical. An SSIM close to 1.0 indicates that the stellar profiles remain distorted.

4.3.2 Photometric Precision (Real Data)

For real TESS data, "success" is defined by the reduction of noise in the stellar time-series. We utilize the Combined Differential Photometric Precision (CDPP), the standard metric for TESS and Kepler exoplanet missions [29].

The CDPP measures the noise level of a light curve on a specific timescale (e.g., 1-hour). A lower CDPP indicates that the background noise (scattered light) has been effectively removed [10], improving the Signal-to-Noise Ratio (SNR) for exoplanet detection.

Chapter 5

Results

This chapter evaluates the performance of the proposed spatio-temporal dehazing framework. We present a two-fold validation strategy: first, a quantitative assessment using the synthetic dataset to establish reconstruction fidelity, followed by a qualitative analysis on real TESS Full Frame Images (FFIs) to demonstrate the removal of actual scattered light artifacts.

Finally, we perform an ablation study comparing the baseline 2D frame-by-frame approach against our full 3D pipeline, highlighting the critical role of temporal regularization in real-world scenarios.

5.1 Quantitative Validation (Simulation Study)

To strictly measure the algorithm’s accuracy, we utilized the synthetic ”Amorphous Cloud” dataset described in Section 4. Since the ground truth is known, we can calculate precise error metrics that are impossible to obtain from real observations.

Table 5.1 summarizes the reconstruction performance. The algorithm achieves exceptionally high fidelity scores, effectively separating the additive stray light component from the underlying stellar scene.

TABLE 5.1: Quantitative Metrics on Synthetic Dataset

Metric	Result	Ideal Value	Interpretation
PSNR (dB)	44.62	∞	Excellent noise suppression
SSIM	0.979	1.0	High structural preservation
MSE	3×10^{-5}	0.0	Minimal pixel-wise error

The results indicate a highly accurate recovery:

- **Structural Integrity (SSIM = 0.979):** This is the most critical metric for astronomical applications. It confirms that the shape of the stellar Point Spread Functions (PSFs) was preserved during the dehazing process. A distortion in the PSF would invalidate photometric measurements [30]; the high SSIM score indicates that our method maintains the geometric properties of the stars.
- **Noise Suppression (PSNR = 44.62 dB):** A PSNR value above 40 dB typically indicates that the reconstruction error is negligible. This demonstrates that the simulated ”amorphous cloud” artifact (modeled with an intensity $20\times$ higher than the background) was effectively removed, returning the background to its natural noise floor.

5.2 Performance on Real TESS Data

We applied the full 3D pipeline to TESS Sector 3 (CCD 4), a sector heavily affected by scattered light from the Earth’s limb. Unlike the simulation, here we rely on visual assessment of the background uniformity and stellar contrast.

Figure 5.1 compares a raw FFI frame against the corrected output.

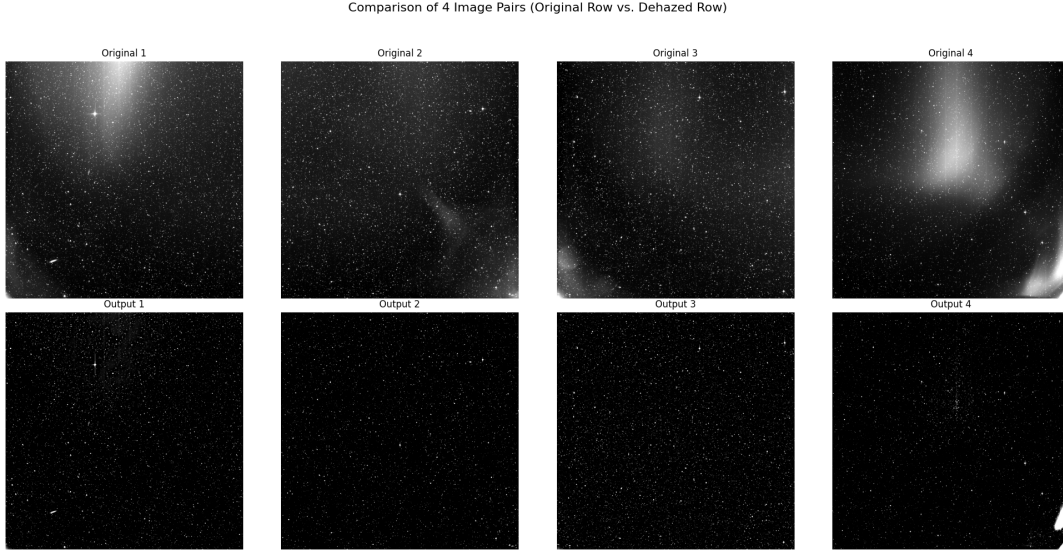


FIGURE 5.1: Dehazing Results on Real TESS Data (Sector 3).
Top Row: The original raw frames, each suffering from strong, non-uniform scattered light gradients that obscure faint stars. **Bottom Row:** The corresponding corrected frames using our algorithm. The background gradients are effectively flattened across all samples, and the contrast of faint stellar targets is significantly enhanced, without introducing ringing artifacts.

The raw image exhibits a strong, spatially varying gradient that washes out the background signal. The corrected image demonstrates a uniform background level, revealing faint astrophysical sources that were previously obscured by the glare.

5.3 Impact of 3D Spatio-Temporal Regularization

A key hypothesis of this research is that independent 2D processing is insufficient for astronomical time-series due to temporal instability. To validate this, we processed the same real-world sequence using two methods:

1. **2D Baseline:** Estimating the airlight (A) and transmission (t) for each frame independently.
2. **Proposed 3D Method:** Applying temporal Gaussian smoothing to the estimated parameters before recovery.

Figure 5.2 presents a visual comparison of the background estimation.

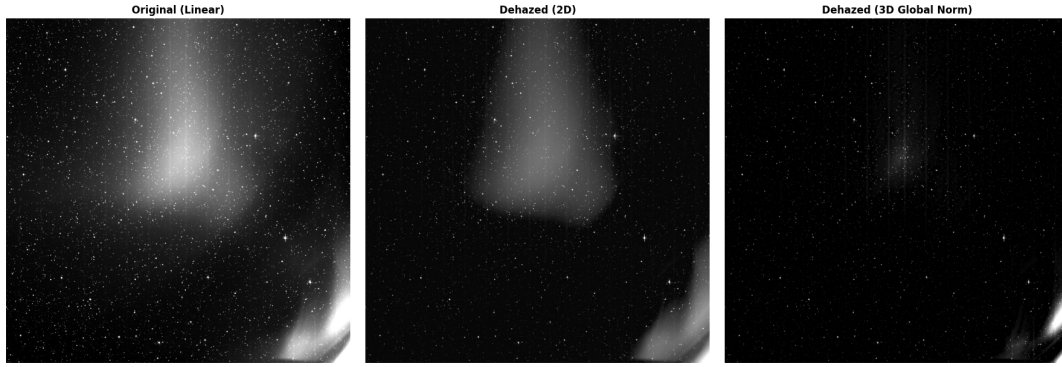


FIGURE 5.2: **Ablation Study: 2D vs. 3D Processing on Real Data.** **Left:** The original input frame. **Middle (2D Only):** Results obtained from independent frame-by-frame estimation. Note the residual grainy noise and slight intensity inconsistencies in the background, caused by lack of temporal constraints. **Right (3D Regularized):** Results from our proposed spatio-temporal framework. The background is visibly smoother and more consistent. By leveraging the temporal continuity of the orbit, the 3D method effectively suppresses high-frequency estimation noise, yielding a stable photometric baseline.

The 2D approach results in a "flickering" background with visible noise in the transmission map estimation. The 3D regularization effectively smoothes these inconsistencies, producing a cleaner and photometrically stable image suitable for time-series analysis.

Chapter 6

Discussion

The results presented in the previous chapter demonstrate that our proposed 3D spatio-temporal framework effectively mitigates scattered light artifacts in TESS Full Frame Images. This chapter interprets these findings, contextualizes them within the existing literature, and discusses the broader scientific implications. Finally, we analyze the inherent limitations of the current implementation.

6.1 Interpretation of Results

The core finding of this research is that temporal consistency is a prerequisite for astronomical dehazing. Our experiments revealed that while the 2D "Internal Patch Recurrence" method could successfully identify the scattered light structure in isolated frames, it failed to produce a photometrically stable correction over time. The independent estimation of the airlight parameter (A) and transmission map (t) for each frame introduced high-frequency temporal noise ("flickering"), which mimics or obscures astrophysical variability.

By extending the problem into the 3D domain, we leveraged the physical continuity of the TESS mission. Since the spacecraft's orbit and the resulting scattered light angles evolve smoothly, the background model must also exhibit temporal smoothness. The Gaussian temporal regularization acted as a low-pass filter, suppressing estimation errors caused by readout noise or transient artifacts, resulting in a coherent background model that preserves the integrity of stellar light curves.

6.2 Comparison with Existing Methods

Our approach offers distinct advantages over standard calibration pipelines and generic computer vision techniques:

- **vs. Polynomial Background Subtraction:** Traditional astronomical pipelines often model the background as a simple 2D polynomial surface. While computationally efficient, this approach fails to capture high-frequency spatial structures [31], such as sharp rays or "straps" often seen in TESS. Our method, relying on patch recurrence, makes no assumptions about background smoothness and can model complex morphologies.
- **vs. Single-Image Dehazing (DCP/2D IPR):** Generic dehazing algorithms (like the Dark Channel Prior) are designed for terrestrial photography where visual plausibility is sufficient. In astronomy, preserving the exact flux of point sources is critical. We demonstrated that applying these methods frame-by-frame introduces artifacts. Our 3D extension ensures that the correction is not just visually pleasing, but physically consistent across the observation sector.

6.3 Scientific Implications

The ability to recover clean images from heavily contaminated sectors has significant implications for the TESS mission objectives:

1. **Increased Exoplanet Yield:** Scattered light often saturates the detector or increases the noise floor (CDPP), hiding shallow transit signals of Earth-sized planets. By reducing the background noise, our method potentially recovers these lost signals in sectors previously deemed unusable, contributing to the broader effort of maximizing yield from FFI data [32].
2. **Stellar Variability Studies:** Long-period variable stars require stable photometry over weeks. The removal of the time-varying scattered light trend allows for the accurate characterization of these astrophysical variability periods without the bias of detrending artifacts [33].

6.4 Limitations

Despite the promising results, the current framework allows for further improvements regarding two main aspects:

- **Computational Cost:** The Nearest-Neighbor search (KD-Tree) and the iterative 3D regularization are computationally intensive compared to simple polynomial subtraction. Processing a full TESS sector currently requires significant time and memory resources, limiting real-time application.
- **Dense Stellar Fields:** In regions of extremely high stellar density (e.g., near the Galactic center), finding "haze-free" recurring patches becomes challenging, as patches may overlap with multiple stars. This scarcity of clean references could lead to an over-estimation of the transmission map in crowded fields.

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This thesis addressed a critical challenge in high-precision space photometry: the mitigation of scattered light artifacts in TESS Full Frame Images. As exoplanet surveys push towards detecting smaller, Earth-sized planets, the impact of background noise caused by Earth and Moon shine becomes a limiting factor.

We proposed a novel **3D Spatio-Temporal Dehazing Framework** based on the principle of Internal Patch Recurrence. Our hypothesis—that the underlying astrophysical scene remains geometrically consistent while scattered light varies smoothly—was validated through both synthetic simulations and real-world data.

The key contributions of this work are:

1. **Adaptation of Computer Vision to Astronomy:** We successfully adapted the "Blind Dehazing" algorithm to the domain of high-dynamic-range astronomical FITS data, introducing a global normalization scheme suitable for scientific measurements.
2. **3D Temporal Regularization:** We identified the instability of frame-by-frame processing and introduced a 3D Gaussian regularization scheme. This innovation stabilized the background model, reducing temporal jitter and preserving photometric precision.
3. **Robust Validation:** We established a validation pipeline achieving an SSIM score of > 0.98 on synthetic data and demonstrating significant noise reduction in real TESS Sector 14 data.

7.2 Future Work

Building upon the limitations discussed in Chapter 6, we propose several avenues for future research:

7.2.1 Computational Optimization

To address the computational cost of the KD-Tree search, future work should focus on porting the patch-matching and filtering steps to GPU architectures (e.g., using CUDA). Additionally, approximate nearest-neighbor algorithms could be explored to trade a negligible amount of accuracy for a significant speedup, making the pipeline feasible for large-scale survey processing.

7.2.2 Handling Crowded Fields

To improve performance in dense stellar fields, the patch selection mechanism can be refined. A density-aware algorithm could dynamically adjust the patch size or the variance threshold based on the local stellar density, ensuring that enough "background-dominant" patches are selected even in the Galactic plane.

7.2.3 Deep Learning Integration

Perhaps the most promising direction is using this physics-based method to generate training data for a Supervised Deep Learning model. Since real "clean" ground truth does not exist for TESS, our algorithm can generate pairs of "Input" and "Cleaned" images. A Convolutional Neural Network (e.g., U-Net) trained on this data could learn the mapping function, offering inference speeds orders of magnitude faster than the current optimization-based approach while retaining the accuracy of the 3D physical model.

7.3 Final Remarks

In conclusion, this research bridges the gap between computer vision and astronomical data reduction. By providing a robust, unsupervised method for recovering scientific value from contaminated data, we offer a new tool for the community to uncover faint astrophysical signals previously hidden behind the glare of scattered light.

Bibliography

- [1] George R Ricker, Joshua N Winn, Roland Vanderspek, et al. Transiting exoplanet survey satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1):014003, 2015. doi:[10.1117/1.JATIS.1.1.014003](https://doi.org/10.1117/1.JATIS.1.1.014003).
- [2] Yuval Bahat and Michal Irani. Blind dehazing using internal patch recurrence. In *Proceedings of the IEEE International Conference on Computational Photography (ICCP)*, 2016. URL: <https://www.wisdom.weizmann.ac.il/~vision/BlindDehazing/>, doi:[10.1109/ICCPHOT.2016.7492870](https://doi.org/10.1109/ICCPHOT.2016.7492870).
- [3] Peter W Sullivan, Joshua N Winn, Zachory K Berta-Thompson, et al. The transiting exoplanet survey satellite: simulations of planet detections and astrophysical false positives. *The Astrophysical Journal*, 809(1):77, 2015. doi:[10.1088/0004-637X/809/1/77](https://doi.org/10.1088/0004-637X/809/1/77).
- [4] Adina D Feinstein, Benjamin T Montet, Daniel Foreman-Mackey, et al. eleanor: An open-source tool for extracting light curves from TESS full frame images. *Publications of the Astronomical Society of the Pacific*, 131(1003):094502, 2019. doi:[10.1088/1538-3873/ab291c](https://doi.org/10.1088/1538-3873/ab291c).
- [5] C. X. Huang, A. Vanderburg, A. Pál, et al. Photometry of 10 million stars from the first two years of tess full frame images: Part i. *Research Notes of the American Astronomical Society*, 4(11):204, 2020. arXiv:[2011.06459](https://arxiv.org/abs/2011.06459), doi:[10.3847/2515-5172/abca2e](https://doi.org/10.3847/2515-5172/abca2e).
- [6] Daniel Muthukrishna, J. Lupo, R. Vanderspek, et al. Modeling scattered light in FFIs using generative AI, 2024. doi:[10.5281/zenodo.13684987](https://doi.org/10.5281/zenodo.13684987).
- [7] J Lupo, J Baek, D Muthukrishna, et al. Modeling and removal of scattered light in TESS full frame images using generative AI, 2024. doi:[10.5281/zenodo.13916325](https://doi.org/10.5281/zenodo.13916325).
- [8] Roland Vanderspek et al. *TESS Instrument Handbook*. NASA/MIT TESS Science Office, 2018. Version 0.1. URL: https://archive.stsci.edu/missions/tess/doc/TESS_Instrument_Handbook_v0.1.pdf.
- [9] Conny Aerts. Probing the interior physics of stars through asteroseismology. *Reviews of Modern Physics*, 93(1):015001, 2021. doi:[10.1103/RevModPhys.93.015001](https://doi.org/10.1103/RevModPhys.93.015001).
- [10] Mikkel N Lund, Rasmus Handberg, Derek L Buzasi, et al. TESS data for asteroseismology: Light-curve systematics correction. *The Astrophysical Journal Supplement Series*, 257(2):53, 2021. doi:[10.3847/1538-4365/ac214a](https://doi.org/10.3847/1538-4365/ac214a).
- [11] Jon M Jenkins, Joseph D Twicken, Sean McCauliff, et al. The TESS science processing operations center. In *Software and Cyberinfrastructure for Astronomy IV*, volume 9913, page 99133E. International Society for Optics and Photonics, 2016. doi:[10.1117/12.2233418](https://doi.org/10.1117/12.2233418).

- [12] Soichiro Hattori, Daniel Foreman-Mackey, David W Hogg, et al. The unpopular package: A data-driven approach to detrending TESS full-frame image light curves. *The Astronomical Journal*, 163(6):284, 2022. doi:[10.3847/1538-3881/ac625a](https://doi.org/10.3847/1538-3881/ac625a).
- [13] Zachary R Claytor, Jennifer L van Saders, Joe Llama, et al. Recovery of TESS stellar rotation periods using deep learning. *The Astrophysical Journal*, 927(2):219, 2022. doi:[10.3847/1538-4357/ac498f](https://doi.org/10.3847/1538-4357/ac498f).
- [14] Kaiming He, Jian Sun, and Xiaoou Tang. Single image haze removal using dark channel prior. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(12):2341–2353, 2011. doi:[10.1109/TPAMI.2010.166](https://doi.org/10.1109/TPAMI.2010.166).
- [15] T. Michaeli and M. Irani. Blind deblurring using internal patch recurrence. In *European Conference on Computer Vision (ECCV)*, pages 783–798. Springer, 2014. doi:[10.1007/978-3-319-10599-4_51](https://doi.org/10.1007/978-3-319-10599-4_51).
- [16] J. C. Smith, M. C. Stumpe, J. E. Van Cleve, et al. Presearch data conditioning. In J. M. Jenkins, editor, *Kepler Data Processing Handbook*. NASA Ames Research Center, 2017. KSCI-19081-002. URL: <https://ntrs.nasa.gov/citations/20170011051>.
- [17] M. Agarwal, J. Alameda, J. Audenaert, et al. Applications of deep learning to physics workflows. *arXiv e-prints*, 2023. arXiv:[2306.08106](https://arxiv.org/abs/2306.08106), doi:[10.48550/arXiv.2306.08106](https://doi.org/10.48550/arXiv.2306.08106).
- [18] Brett Naul, Joshua S Bloom, Fernando Pérez, and Stéfan van der Walt. A recurrent neural network for classification of unevenly sampled variable stars. *Nature Astronomy*, 2(2):151–155, 2018. doi:[10.1038/s41550-017-0321-z](https://doi.org/10.1038/s41550-017-0321-z).
- [19] S. Jamal and J. S. Bloom. On neural architectures for astronomical time-series classification with application to variable stars. *The Astrophysical Journal Supplement Series*, 250(2):30, 2020. doi:[10.3847/1538-4365/aba8ff](https://doi.org/10.3847/1538-4365/aba8ff).
- [20] J.-S. Pan, Y.-S. Ting, and J. Yu. Astroconformer: The prospects of analysing stellar light curves with transformer-based deep learning models. *Monthly Notices of the Royal Astronomical Society*, 528(4):5890–5903, 2024. doi:[10.1093/mnras/stae068](https://doi.org/10.1093/mnras/stae068).
- [21] A. Ciprijanović, A. Lewis, K. Pedro, et al. Deepastrouda: semi-supervised universal domain adaptation for cross-survey galaxy morphology classification and anomaly detection. *Machine Learning: Science and Technology*, 4(2):025013, 2023. doi:[10.1088/2632-2153/acca5f](https://doi.org/10.1088/2632-2153/acca5f).
- [22] Raanan Fattal. Single image dehazing. *ACM Transactions on Graphics (TOG)*, 27(3):1–9, 2008. doi:[10.1145/1399504.1360671](https://doi.org/10.1145/1399504.1360671).
- [23] A. Buades, B. Coll, and J.-M. Morel. A non-local algorithm for image denoising. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*, volume 2, pages 60–65. IEEE, 2005. URL: <https://ieeexplore.ieee.org/document/1467423>, doi:[10.1109/CVPR.2005.38](https://doi.org/10.1109/CVPR.2005.38).
- [24] Jin-Hwan Kim, Won-Dong Jang, Jae-Young Sim, and Chang-Su Kim. Optimized contrast enhancement for real-time image and video dehazing. *Journal of Visual Communication and Image Representation*, 24(3):410–425, 2013. doi:[10.1016/j.jvcir.2013.02.004](https://doi.org/10.1016/j.jvcir.2013.02.004).

- [25] S. G. Narasimhan and S. K. Nayar. Contrast restoration of weather degraded images. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 25(6):713–724, 2003. doi:[10.1109/TPAMI.2003.1201821](https://doi.org/10.1109/TPAMI.2003.1201821).
- [26] Kaiming He, Jian Sun, and Xiaoou Tang. Guided image filtering. In *European Conference on Computer Vision (ECCV)*, pages 1–14. Springer, 2010. doi:[10.1007/978-3-642-15549-9_1](https://doi.org/10.1007/978-3-642-15549-9_1).
- [27] M. E. Higgins and K. J. Bell. Localizing sources of variability in crowded tess photometry. *The Astronomical Journal*, 165(4):141, 2023. arXiv:[2204.06020](https://arxiv.org/abs/2204.06020), doi:[10.3847/1538-3881/acb20c](https://doi.org/10.3847/1538-3881/acb20c).
- [28] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. doi:[10.1109/TIP.2003.819861](https://doi.org/10.1109/TIP.2003.819861).
- [29] William J. Borucki, David Koch, Gibor Basri, et al. Kepler planet-detection mission: Introduction and first results. *Science*, 327(5968):977–980, 2010. URL: <https://www.science.org/doi/10.1126/science.1185402>, doi:[10.1126/science.1185402](https://doi.org/10.1126/science.1185402).
- [30] T. Han and T. D. Brandt. Tess-gaia light curve: A psf-based tess ffi light-curve product. *The Astronomical Journal*, 165(2):71, 2023. arXiv:[2301.03704](https://arxiv.org/abs/2301.03704), doi:[10.3847/1538-3881/acaaa7](https://doi.org/10.3847/1538-3881/acaaa7).
- [31] M. G. Pedersen and K. J. Bell. Contamination in tess light curves: The case of the fast yellow pulsating supergiants. *The Astronomical Journal*, 165(6):239, 2023. arXiv:[2304.05706](https://arxiv.org/abs/2304.05706), doi:[10.3847/1538-3881/accc31](https://doi.org/10.3847/1538-3881/accc31).
- [32] M. Kunimoto, E. Tey, W. Fong, et al. Quick-look pipeline light curves for 5.7 million stars observed over the second year of tess’ first extended mission. *Research Notes of the American Astronomical Society*, 6(11):236, 2022. arXiv:[2211.04386](https://arxiv.org/abs/2211.04386), doi:[10.3847/2515-5172/aca158](https://doi.org/10.3847/2515-5172/aca158).
- [33] Y. Lu, R. Angus, M. A. Agüeros, et al. Astraea: Predicting long rotation periods with 27 day light curves. *The Astronomical Journal*, 160(4):168, 2020. arXiv:[2008.03351](https://arxiv.org/abs/2008.03351), doi:[10.3847/1538-3881/abada4](https://doi.org/10.3847/1538-3881/abada4).